
Pattern Arithmetic: Phase-Colored Geometric Framework for High-Density Information via Unit Probability Spheres

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Abstract

Traditional matrix representations of continuous state spaces suffer from severe structural limitations: rigid coordinate axis dependencies, exponential spatial scaling costs ($O(N^3)$), and the destructive overwrite of overlapping signals at identical spatial coordinates. To overcome these constraints, this paper introduces *Pattern Arithmetic*, a continuous geometric framework that models multi-variable information states using state vectors on a Unit Probability Sphere $\mathcal{S} \subset \mathbb{R}^3$, augmented by an internal phase-color degree of freedom $\chi \in [0, 2\pi)$. By mapping continuous probability distributions over vector fields and treating color as an independent phase factor, overlapping states occupy identical spatial coordinates without loss of information density. We establish four foundational operational axioms, including Information Idempotence ($1 + 1 = 1$), Incremental Subsumption ($A + A.01 = A.01$), and Phase Interference ($A - A = 0$). Finally, we demonstrate how this pattern-based algebra provides a parameter-efficient alternative for high-density latent storage, signal unmixing, and physical field modeling.

1 Introduction

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1.1 The Limitations of Discretized Matrix Grids

- **Memory Scaling Bottlenecks:** $O(N^3)$ spatial growth in 3D discretized voxel grids.
- **Signal Identity Destruction:** Overwriting of overlapping signals when mapping multiple continuous variables to identical spatial coordinates.

1.2 The Phase-Colored Unit Probability Sphere Framework

- **State Vector Definition:** $v = (r, \theta, \phi, \chi)$ where r is the magnitude likelihood ($0 \leq r \leq 1$), (θ, ϕ) are spherical coordinates defining spatial orientation, and χ is the internal phase-color degree of freedom ($0 \leq \chi < 2\pi$).
- **Continuous Probability Distribution Mapping:** Each state vector represents a continuous probability distribution over the unit sphere $\mathcal{S}$, allowing for high-density information representation without coordinate grid expansion.

1.3 Core Contributions & High-Level Intuition

- **High-Density Information Superposition:** Storing overlapping state vectors without expanding coordinate grids.
- **Parameter Efficiency:** Continuous field representation without discrete voxel bounds.

2 Axiomatic Foundations Of Pattern Arithmetic

2.1 Axiom 1: The Normalized Unit Truth State ($S = 1$)

The global state space is strictly bounded to unit integral likelihood:

$$\iiint_S P(x, y, z) dV = 1.0 \quad (1)$$

where $S = 0$ represents the Null State ($\|\vec{v}\| = 0$, Black) and $S = \infty$ represents Infinite Superposition (White).

2.2 Axiom 2: Information Idempotence & Non-Redundancy ($A + A = A$)

Superimposing identical information patterns yields no state expansion. Upon normalization over intensity:

$$A \oplus A = \frac{A + A}{\|A + A\|} = A \quad (2)$$

2.3 Axiom 3: Incremental State Subsumption ($A + A.01 = A.01$)

When an updated prior A_{new} subsumes an earlier state A_{old} , state update dynamics shift to the refined wave packet:

$$\lim_{\delta \rightarrow 0} (A + A_{\delta}) = A_{\delta} \quad (3)$$

2.4 Axiom 4: Phase-Dependent Vector Interference

Vector combination is governed by the relative phase difference $\Delta\chi = \chi_1 - \chi_2$:

$$A \pm B = \sqrt{\|A\|^2 + \|B\|^2 + 2\|A\|\|B\|\cos(\Delta\chi)} \quad (4)$$

3 Operational Algebra & Numerical Demonstration

3.1 Vector Addition & Phase Blending

Using Equation (4) from Axiom 2.4, combining Red Vector (0°) and Blue Vector (120°) at coordinate $(0.6, 0.8)$ results in deterministic phase blending ($\chi = 66.6^\circ$).

3.2 Rotational Transposition (Multiplication) & Unmixing (Division)

- **Multiplication ($\times$):** State transformation via Lie group rotation matrix $\mathbf{R} \in \text{SO}(3)$.
- **Division ($\div$):** Deconvolution and signal unmixing via inverse transformation $\mathbf{R}^{-1}$ or anti-phase isolation.

3.3 Computational Efficiency & Spectral Compression

- Mitigating $O(N^3)$ voxel grid costs using Spherical Harmonics (Y_l^m) coefficient arrays $O(L^2)$.

4 Real-World Physical & Biological Models

4.1 Forest Ecosystem Dynamics

- Modeling flora, fauna, soil, and moisture resource completion via vector addition ($A + B = AB$).

4.2 Targeted Viral/Pathogen Cancellation (Selective Subtraction)

- Isolating disease signatures and applying 180° anti-phase mirror vectors to achieve zero-state equilibrium ($A - B = 0$).

4.3 Atmospheric Storm Systems (Phase Shifts and Shear Turbulence)

- In-phase wave coherence vs. minor phase shift (0.01) driving localized vortices and trajectory shifts.

5 Discussion & Related Work

5.1 Comparison to Quantum Bloch Spheres and Hilbert Spaces

5.2 Integration with Machine Learning Latent Embeddings (Cosine Similarity)

5.3 Hardware Realization: Optical Phase Computing vs. Digital GPUs

6 Conclusion & Future Directions